B. Patients and Caregivers

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

The Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (ASTP)/CMS should improve the development environment through funding the development of innovative, open-source, gold-standard applications and patient-facing tools that could be published as a template for other developers. These tools should leverage mature interoperability standards (e.g., United States Core Data for Interoperability [USCDI], Fast Healthcare Interoperability Resources [FHIR]), be built to the most rigorous standards (not limited to current feasibility—rather set the bar as high as possible, signaling clearly defined, aspirational targets). Development of such innovative applications is difficult in the private/non-profit sector as the resources and expertise necessary rarely reside within a single company or organization. Development of open access tools that become the gold standard for other private developers can only be accomplished through "coopetition" funded by the government. Blue Button 2.0 also serves as an excellent model to emulate.

The development of these applications should prioritize end-user input at all stages of development, providing substantive feedback on basic navigation as well as:

1. Address questions regarding their need for information from their health plan (network capacity, eligibility, prior authorization requirements, quality ratings, cost, and accessibility, scheduling, specialty, substance use and mental health support).
2. Address concerns regarding privacy regarding how the information will be used to improve their health or the health of those under their care. Presentation of this information should conform to legal requirements but be transparent about other potential uses.
3. Address issues caregivers face in managing applications for people other than themselves. This should provide useful and understandable information about permissions and access that are not generally a priority in the private app dev market.

PC-1 a. What are the top things you would like to be able to do for you or your loved ones' health that can be enabled by digital health products?

1. Care planning: Ability of the patient/caregiver to create, modify, integrate, and/or correct issues in the care plan in the longitudinal record. Ideally, a care plan is owned by the patient/caregiver tracking their goals and outcomes. Currently, multiple care plans may be created by different providers/care teams without the ability to harmonize with each other or even follow a hierarchical structure for updating with patient/caregiver tracking updates.
2. Value to the patient: The patient must perceive the value of any digital health product and trust that sharing information directly benefits them. Patients/caregivers must have clear incentives to engage through digital health products. To increase active participation, the ability to view and request changes/education/support to the longitudinal record is

critical, particularly when a care plan is executed/updated and treated as if owned by the patient.

PC-1 b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

The top priority would be navigation and coordination of healthcare delivery. Scheduling, transportation, food, shelter, and access to primary care, specialty care, care management, mental health services, and substance use disorder treatment currently are all very segmented across the various care settings. A professional navigator that understands the issues within each care network and can assist patients in coordinating their healthcare journey will lead to higher patient satisfaction as well as better outcomes.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

Access is highly subjective. While access to current health portals is relatively easy to establish, the ability to navigate once inside the portal is very difficult. These portals are not designed with the end-user in mind, whether that person is a patient or a provider. The information is segmented, difficult to search across the various areas of the record (e.g., current medications, past medical history, current conditions) and is not easily updated/corrected by the patient. Some platforms do not even make patient-entered information viewable by the person's clinical team, no matter the veracity or completeness of the information entered.

PC-2 a. If so, what are some examples of benefits it has provided?

Electronic health record (EHR) portals can be very useful for managing one's own health, yet they don't currently support continuity with providers. EHR portals like Epic's Care Everywhere can present information on out-of-network providers and can be useful for scheduling visits, conducting medication management, and looking up immunization information.

PC-2 b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

A single access point with a linking identifier would be ideal. The access point would be based on human-centered design principles informed by substantial user experience inputs. End-user centered design in product development is essential for success.

A critical function of a patient portal would be to effectively coordinate and manage care plans (across multiple providers) to facilitate easy navigation by patients and shared decision making around steps in care, recovery, rehabilitation, and maintenance to reach good health outcomes.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

The patient-focused Kidney Choices app, developed by IPRO with CMS input, is available on the web, the Apple App Store and Google Play. The app has two tracks:

- CKD on Track allows patients to access their Medicare Blue Button 2.0 data to track laboratory monitoring, provider visits, and selective drug dispensing, and to provide notifications of when the next lab, provider visit, or drug refill is due. The app also offers patients the ability to communicate due dates with their healthcare providers electronically.
- Kidney Treatment Choices allows patients to explore each end-stage renal disease treatment option: hemodialysis, peritoneal dialysis, transplant, and conservative care. While patients explore options, they can simultaneously create a list of questions about each option; these questions can then be transmitted electronically to their healthcare provider for shared decision making.

PC-4. What features are missing from apps you use or that you are aware of today?

PC-4 a. What apps should exist but do not yet? Why do you believe they do not exist yet?

Development of innovative, open-source applications is exceptionally difficult in the private/non-profit sector as the resources and expertise necessary rarely reside within a single company or organization. Development of open access tools that become the gold standard for other private developers can only be accomplished through "coopetition" funded by the government.

The development of these applications should prioritize end-user input at all stages of development, providing substantive feedback on basic navigation. The consumer healthcare app industry is lacking priority on privacy and security. There isn't a sense of ownership of the information with health apps. Health apps should be as easy as financial banking apps which enable bringing information together from multiple sources and the ability to move/invest/track/report/self-educate/self-guide/act/record/receive alerts with a high level of confidence that the information is protected and secure at all stages. Additionally, there is a lack of trust in the technological realm with highly sensitive and confidential information.

PC-4 b. What set of workflows do you believe CMS is uniquely positioned to offer?

ASTP/CMS should support the development of innovative, open-source, gold-standard applications and patient-facing tools that can be published as a template for other developers. These tools should leverage mature interoperability standards (e.g., USCDI, FHIR), be built to the most rigorous standards (not limited to current feasibility—rather set the bar as high as possible, signaling clearly defined, aspirational targets). ASTP and CMS should continue to support consensus-based interoperability standards development and maintenance. These standards should not be developed to accommodate the least common denominator; rather, they should set goals that align regulatory requirements with quality improvement targets. By continuing support for Health Level Seven (HL7) and USCDI standards, as well as promoting the use of standard terminologies (e.g., Systematized Nomenclature of Medicine [SNOMED], RxNorm, Logical Observation Identifiers Names and Codes [LOINC]) referenced by those standards will aid the digital quality ecosystem in meeting CMS' modernization and digitalization goals.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

Consumer focused dissemination strategy: Widely disseminate the benefits of a digital health system versus dependence on EHRs, etc. Consumer apps have transformed other industry experiences with a well-defined gain for the consumer. Lead the transformation toward the public embracing ownership of their longitudinal health record where care plans or wellbeing plans can be centralized.

Incentives: There are a limited number of personal health programs that incentivize tracking of health status and monitoring progress towards goals. There is a reasonable amount of data indicating that the value of the incentive does not need to be very high, just sufficient to keep the person engaged in order to realize the eventual benefit from participation.

PC-5 a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)?

Certification: Develop a certification program based upon standard criteria for applications ensuring they meet rigorous standards for useability, compatibility with other apps, and interoperability of the data generated by the digital health product. Ensure that the certification process includes a component that evaluates the level of patient/caregiver experience in the design and testing process. Include more than one level of criteria such that apps have the option of becoming designated as a top performer against rigorous certification criteria.

PC-5 a (*cont'd.*). What criteria should be used if there is a review process?

1. Adherence to human-centered design principles:
    a. The design is based upon an explicit understanding of users, tasks, and environments.
    b. Users are involved throughout design and development.
    c. The design is driven and refined by user-centered evaluation.
    d. The process is iterative.
    e. The design addresses the whole user experience.
    f. The design team includes multidisciplinary skills and perspectives.
2. Interoperability/InterOptimability:
    a. Interoperability: the ability of different healthcare information systems to communicate and exchange data in a coordinated manner.
    b. InterOptimability: a conceptual framework and practical methodology that helps organizations focus on integrating and optimizing systems to improve operations and service delivery. It offers a pragmatic approach to leverage organizational investments in integration to produce more insights, make better decisions, and guide development of holistic, person-centered and effective systems of care.
3. Compatibility with multiple common data models/structures:
    a. While USCDI and FHIR are of primary interest in evaluating applications for conformance, it would be short sighted not to acknowledge the useability of information best suited for evidence generation and research. This will allow apps in the digital health environment to support the latest and best evidence-based

practices whether or not they are directly related to, or derived from, electronic health records.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

It is essential to incorporate user experience design principles into a process of creating digital health products and services that solve a specific user problem (e.g., limited experience with digital tools/services), while ensuring that the proposed solution is both easy and enjoyable to use. Applications must be fully functional on all mobile platforms and integrate seamlessly with more traditional person-to-person interaction.

Support for deployment-counseling, education, and promotion by providers requires customized messaging. Also, frequent re-evaluation of the user experience and feedback is necessary. It must not only update for technical reasons, but also address end user feedback, building trust to help patients and caregivers use digital health products throughout the wellbeing journey, and providing information capture through frequent surveys and reporting on risk factors and care plans.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

Effective measurement: CMS should focus on developing and deploying a series of measurement protocols that are tailored to function effectively in a digital healthcare environment. These must approach measurement holistically and include both patient-reported and clinical outcomes together to assess impact.

This must be accomplished using a person-centric approach which accounts for, and puts in context, each patient's personal goals as well as the clinical results from the care team perspective. Success should be measured over time as patients progress toward the ideal state and should not be focused solely on measuring individual processes as discrete, unconnected variables. Focusing on measuring health status through validated patient-reported outcome instruments captures how people perceive their health and is a predictor of important health outcomes including mortality, morbidity, and functional status.

The patient perspective is increasingly a priority as quality strategies orient more around patient-centeredness. A person-centered measurement model that evaluates how well clinicians manage risk and health outcomes across their entire patient population with respect to individual clinical biomarkers and patient characteristics is the innovative solution to the disruption caused by performance measurement.

Measure scores should reflect successful management that is sensitive to changes in *individual* patient risk factors over time as this estimates expected outcomes under different treatment scenarios. Focusing on meaningful, incremental risk reduction helps clinicians optimize treatment of existing conditions while concurrently improving the overall health of their patients by supporting decision-making interactions oriented around care expectations and personal goals. New measurement strategies must promote a culture of continuous improvement,

systematically identifying opportunities to enhance the quality and efficiency of care delivery while maximizing the likelihood of achieving positive outcomes for patients.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

PC-8 a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

Data dumping from patient apps is not useful to anyone (especially the provider) as the volume of data overwhelms the record. Oftentimes information for a patient's encounter becomes duplicated repeatedly within the medical record. Additionally, other vital information stemming from other patient visits is not seamlessly integrated into the patient's main medical record, while the EHR's data prioritizes data created within its system, and deprioritizes externally obtained data.

While patient data should be able to be directly integrated into their medical record via Substitutable Medical Applications, Reusable Technologies (SMART) on FHIR applications, the apps need to present the data with sufficient specificity to be meaningful to the clinician's reasoning as well as summarized in such a way to only highlight the most relevant and important factors for that particular moment of clinical decision making. There is very good ongoing work in the standards communities (e.g., minimal Common Oncology Data Elements [mCODE] Implementation Guide and CardX Hypertension Management) defining measurement approaches in such a way to make patient app data dumping unnecessary. Highlighting the most important outcomes (whether they are intermediate or terminal) provides structure for app developers to ensure the most relevant information is available to inform clinical decision making.

It is not always just about the data itself—the provenance and data quality inform its use. The value comes from integrating data from multiple sources with purposeful guidance on end user expectations for its use in any particular context. To act on any information, it must be relevant to patients/caregivers for decision-making on needed services, their cost, and clinical provider experience.

PC-8 b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

Each data source accessed by a digital health application or tool is valuable for a specific context. For example, an app that allows a person access to their immunization data in the Immunization Information System (IIS) can be very valuable for everything from keeping up to date with vaccine schedules to meeting school/work vaccination information requirements.

Setting personal health goals is an important aspect of living well. Apps that can assist people in setting goals based on what is most important to them, and that use a model for tracking progress that is sensitive to modest success—while still being clinically relevant—will encourage continued use. Connecting non-clinical goals to clinical recommendations and reconciling what approach would be most beneficial to any individual would be very valuable.

Bulk FHIR application programming interfaces (APIs), alignment of data exchange for quality measurement, and FHIR capabilities through FHIR resources should be encouraged to capitalize on use of digital health products bringing data together that is evidence-based and can be spread and scaled across organizational use cases.

PC-8 c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

The link between the standards most utilized for research and evidence generation and those supporting interoperability of clinical information is still an area that needs much more support and input from multiple stakeholders. Each standard has its own Common Data Model (CDM) and tooling associated with it with little ability to easily move between them. There are several technological solutions to this problem; however, there has not been a systematic evaluation of the technology that would enhance stakeholder trust in the process or the outcomes. More collaboration is needed on use cases, problem solving, knowledge sharing, and pilot testing of the cross-standard functionality.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

IPRO has had a great experience working with the Blue Button 2.0 team from its inception, in working through technical challenges that have ultimately led to an excellent claims API model today. We are one of the organizations that remains connected to the API and is serving data through our Kidney Choices patient application.

Cross-collaboration can bring in public health data and CAHPS data and integrate clinical and functional status data. Survey data can be collected that captures additional information that can get at risk factors and engage the patient in risk factor profiles knowing this changes with goal setting and acquisition of new data.

PC-9 a. What difficulties are there in accessing or utilizing these data sources today?

There is a lack of commitment to a longitudinal wellbeing record and focus on care plan to activate the patient. Patients need to be engaged in their health journey throughout their lifespan and provide more data via reliable/valid patient surveys (patient-reported outcomes). Capture of risk factors, goal setting, care plan action, and updates at regular intervals are necessary to get at actionable and meaningful information that move outcomes.

PC-9 c. Is there non-CMS data that should be included in the API?

Public health data, CAHPS data, clinical process/outcome data, functional status data, and social determinants of health data should be included. Most of these listed need to be valid/reliable tools to capture at key intervals across the lifespan.

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world? It

provides consistency in policies, approaches for electronic exchange and promotes patient access and control of their health information.

PC-10 a. Please provide specific examples.

The link below to L.A. Care is a very good example of how data exchange should work. https://www.hcinnovationgroup.com/interoperability-hie/fast-healthcare-interoperability-resources-fhir/news/55289555/la-care-using-fhir-to-send-claims-encounter-and-clinical-data-to-lanes-qhio

PC-10 c. What use cases could have a significant impact if implemented through TEFCA?

If CMS was able to learn from the L.A Care use case, and build upon related activities such as San Diego's Community Information Exchange for social determinants of health, these efforts would improve actionable data sharing.

PC-10 d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

HL7 FHIR

PC-10 e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

PCORnet, Observational Medical Outcomes Partnership (OMOP)

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

The HIEs allow comparison shopping of health plans at enrollment time through use of a quality rating system in the marketplace.

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

CMS can prominently disseminate and display ways for reporting and the process for action and resolution. Reporting complaints would advance the impact of data exchange and trust, if addressed with a timely response or mechanism to track the complaint.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

PC-14 a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

There is a lack of awareness and/or trust. Suggestions include:

- Initiate a massive campaign, making it easy to sign up and ask questions.
- Set expectations/recommendations for active use and participation.

- Make the preferred choice the easy choice, by inserting these tools into the patient's workflow.

PC-14 b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

There are several benefits such as improved access, enhanced security, and greater control over one's health data. A huge benefit is reduced error, empowering activation, and streamlined processes. The speed to access reduces paperwork and wait times. Guards against identity theft and digital wallets and other tools can be used to securely store, manage, and share health data. Patients are more informed and ready for shared decision making. Sharing data easily and securely improves coordinated care.

PC-14 c. What are the potential downsides?

Downsides include misuse by others (which affects privacy) and data breaches; yet financial information faces the same perils. CMS should consider learning from security protocols in the financing industry, focusing on National Institute of Standards and Technology (NIST) security protocols.

C. Providers

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

Focus on awareness of the demonstrated value and success stories. Providers need assurance of availability and sustainability of digital health products. Cost and ongoing implementation support are also major factors affecting adoption of digital health technology. Address barriers to use as provided below.

PR-1 a. What are the current obstacles?

IPRO has deep experience in advancing healthcare quality improvement in rural areas, including optimizing health information technology. IPRO has worked with providers who have encountered all the obstacles/barriers in leveraging, identified in the evidence below.

Digital tools improve health outcomes for individuals with limited healthcare access in rural areas. Barriers to use include absent or inadequate access to technology and high-speed internet; limited patient digital familiarity, literacy and education, and training; limited patient health literacy; lack of patient trust in digital tools; social determinants of health; and health-related social needs barriers. (Maita KC, Maniaci MJ, Haider CR, Avila FR, Torres-Guzman RA, Borna S, Lunde JJ, Coffey JD, Demaerschalk BM, Forte AJ. The Impact of Digital Health Solutions on Bridging the Health Care Gap in Rural Areas: A Scoping Review. Perm J. 2024 Sep 16;28(3):130-143. doi: 10.7812/TPP/23.134. Epub 2024 Aug 13. PMID: 39135461; PMCID: PMC11404635.)

Commonly reported barriers to digital health technology use center on product complexity and reliability, awareness of resources, lack of trust, and cost. Effective digital health technology use is becoming increasingly important. There may be a need to prioritize and support people with lower levels of digital health literacy. (Jongebloed H, Anderson K, Winter N, Nguyen L, Huggins CE, Savira F, Cooper P, Yuen E, Peeters A, Rasmussen B, Reddy S, Crowe S, Bhoyroo R, Muhammad I, Ugalde A. The digital divide in rural and regional communities: a survey on the use of digital health technology and implications for supporting technology use. BMC Res Notes. 2024 Mar 28;17(1):90. doi: 10.1186/s13104-024-06687-x. PMID: 38549176; PMCID: PMC10976777.)

PR-1 b. What information should providers share with patients when using digital products in the provision of their care?

Providers should share the following information:

- Clear indication of why the digital product is being considered for use (include diagnosis for use), benefits (why the device is better than usual care), and the potential risks associated with use.
- Patient expectations if the device indicates an abnormality (e.g., out of range results/alerts) and provider response (communication methods, timeliness, action/next steps).

- Point of contact(s) for the patient for any device or device-related clinical questions along with multiple communication methods (phone, email, etc.).
- Information regarding how care partners can access or assist with the device.
- Information on how patient identity and health information, including sensitive information (mental health, HIV status), is secured or shared, including if patient data is aggregated for artificial intelligence (AI) use or if AI is providing health information/monitoring/diagnostics. (Light Collective - No Aggregation Without Representation).

PR-1 c. What responsibilities do providers have when recommending use of a digital product by a patient?

Providers must perform due diligence on the digital device or product, which includes the following assurances prior to use by a patient:

- Provider institutions offer only digital devices/products that have been evaluated using a peer-reviewed evidence-based framework (accessibility/usability, subject/content quality, clinical effectiveness, data privacy/security, engagement). Organizations should provide evaluation and oversight of these products and include clinician education to improve uptake. (Mescher, T., Hacker, R.L., Martinez, L.A. et al. Mobile Health Apps: Guidance for Evaluation and Implementation by Healthcare Workers. J. technol. behav. sci. (2024). https://doi.org/10.1007/s41347-024-00441-7).
- Digital devices/products must be clinically evidence based, be FDA approved if applicable, and ensure patient data security, privacy, and cybersecurity protections.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

The landscape of measurement has undergone significant transformations in the past three decades. While the foundational principles and premises guiding the development of quality measures have remained relatively constant, the context within which these measures operate has witnessed substantial changes. As we navigate the complexities of modern healthcare, there is a compelling need to reassess traditional approaches and embrace innovative methodologies that can capture the essence of quality in healthcare delivery.

A major barrier to success is the continued assumption that the process of digitalization alone will bring relief. A digital revolution should be dramatic and bring wide-reaching change which has not happened in the digital healthcare ecosystem. CMS is perfectly placed to "Break the Cycle" that has kept digital quality measurement from effectively modernizing. Supporting healthcare organizations with the transition to digital requires doubling down on resources during the transition phase due to the need to simultaneously run the traditional reports alongside the new methods to ensure no loss of data, revenue, or overall confidence in the process.

Measure Once…Report Twice. Effective use of data generated during the normal course of care for measuring quality AND improving healthcare is essential for success. Eliminating manual extraction/abstraction review streamlines the process, but requires introduction of new AI tools

like natural language processing (NLP) and other machine learning algorithms which have the ability to derive information from unstructured data, and bridge much of the performance gap between human review and structured data collection.

Leverage the benefits that digitalization presents to reduce the manual work to generate measure information. Shift focus to electronic clinical quality measures (eCQMs)/digital quality measures (dQMs) that not only provide useful, timely, and meaningful information but also augment the "Measure Specific" tasks. Quality reporting-specific actions can easily be safely automated within the dQM specification to improve efficiency and reliability, reducing reliance on human-mediated processes unique to measure reporting. Much of the criticism surrounding EHR implementation involves provider abrasion. Clinicians have fundamentally altered their workflow to capture key data elements in structured fields for quality reporting, shifting the focus of their attention from the patient to their EHR systems. Adopters of NLP will be able to invest limited human resources in training/refining these algorithms and performing quality assurance over-read. Organizations can then effectively leverage human clinical expertise to accomplish more diverse reviews, while decreasing the monotony of the review task for critical clinical experts. Many providers continue to document in freeform progress and Subjective, Objective, Assessment, and Plan (SOAP) notes. Some encounters are simply not conducive to structured formats. NLP is an ideal utility to capture data from unstructured text and introduce more flexibility into documentation requirements without losing information for quality reporting.

Improving the value of the information generated by quality measures has always been a challenging task. Finding ways to make information more meaningful for quality improvement means assessment must become more prospective, strategically placing informational triggers that guide the patient-provider interaction toward evidence-based decisions. Quality reporting programs that strive for a more personalized approach will struggle to achieve the aims of the CMS strategy unless measurement mechanisms are better aligned with CMS' value-based goals. In this context, effective use of information technology becomes essential for new assessment methods to succeed, augmenting human-mediated decision processes rather than replacing them. Advancing quality measurement methods goes beyond simply harnessing technology. It requires a fundamental shift toward including more flexible quality assessment models that can elucidate the relationship between patient preferences and improving individual patient experiences. Leveraging advanced information technology to identify and prioritize the needs and preferences of patients enables healthcare organizations to deliver more personalized and holistic care tailored to the unique circumstances of every individual's healthcare situation.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)?

It is essential to modernize healthcare and facilitate seamless interoperability between different systems even if different storage methods are used. Healthcare providers need access to the most complete and up-to-date patient information in order to make informed decisions, leading to better patient outcomes.

Please address all of the following:

PR-3 a. Current challenges in accessing different data formats.

Current challenges include narrative, unstructured data formats, or locked data inside PDFs; and information blocking of specialty information (behavioral health, oncology).

PR-3 b. Impact on patient care quality.

Complete and timely data allows healthcare professionals to make more informed decisions, enhance efficiency with workflow, reduce administrative burden, and focus on patient care.

PR-3 c. Technical barriers to full data accessibility.

Health systems rely primarily on vendors to provide their interoperability and data integration needs. This may limit access to other data that the EHR vendor discourages from integration. Providers should be made aware of benefits/pitfalls of FHIR servers.

PR-3 d. Cost or privacy implications of making all data formats interoperable.

Cost is a major factor in adoption rates for digital health technology. Solutions are available including middleware and data translation tools; however, awareness of such options is often limited. Adoption lags can be mitigated in a stepwise fashion with one effort serving as a catalyst to incrementally move away from legacy systems when the organization is ready to do so.

PR-3 e. Priority level compared to other interoperability needs.

This is crucial for modernizing healthcare, reducing errors, being informed about care, and reducing inefficiencies in processes. Adoption of FHIR, focusing on semantic interoperability and patient data accessibility, will result in better patient care and efficient healthcare operations.

2. Data Exchange

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

Encourage use of tools that maximize benefits of provenance while maintaining security, compliance, and operational efficiency. Tools that support data quality, catalogue data sources, support metadata, and the work/process flow (e.g., Business Process Management Plus [BPM+]) bring value to culled and integrated data from multiple sources with purposeful guidance on interpretation.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

Make development of new, innovative digital quality measurement strategies a priority to set a "gold standard' for quality assurance. Developing assessment strategies that are sensitive to successful, incremental progress toward healthcare goals not only removes the documentation

burden associated with quality reporting but also incentivizes providers to document meaningful data in a person's records that is valuable for the person's ongoing care management. This should be combined with the integration of patient input through apps directly linked to medical records that do not just "dump" patient reported data into the record but follow the latest standards for clinically meaningful presentations of the information alongside the clinician-entered data.

PR-8 b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

dQMS require effective interoperability of healthcare systems to function. dQMs offer a whole world of possibilities; however, without re-configuration to operate effectively in a 21st century digital healthcare data environment where aggregation at scale allows for comprehensive assessment of patients' healthcare journeys around their network, efforts aimed at improving data quality, gap analysis, and reduced administrative burden will never be fully successful. This requires the de novo development of quality measures that 1) are person-centered, 2) are focused on a range of outcomes that change over time, 3) use the data already in the records to assess progress toward a series of health-related goals, and 4) are focused on identifying opportunities to adjust care planning instead of punitive, retrospective approaches that have no value outside of performance reporting. The digital exchange of data for quality measures needs to be re-aligned and thoroughly tested. Many services are currently duplicated across settings to assess performance, creating additional burden through inefficiency. The best solution is grounded in disciplined implementation of best practices for provenance and data quality.

PR-8 c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

The main limitation with HIEs and registries supporting quality measurement is there is no standard format for data quality assessment (DQA). Current DQA dashboards are tailored to a very specific audience and tend to be bespoken for every use case. CMS should set a base standard for data quality frameworks and provide significant guidance about data quality expectations (based on extensive end-user feedback).

It is essential that both HIEs and registries can report their capabilities regarding common data model compatibility and comprehensive data fitness standards. This extends beyond just FHIR as many registries should be able to support PCORnet and OMOP to enable their data assets use in essential research generating the next generation of evidence and clinical guidance.

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

PR-9 a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

- Providers would need to know why the digital ID is being considered for use, benefits, and the potential risks associated with use.
- Providers will need support navigating patient expectations to functional and content features, and what should be addressed by the provider or by others.
- They need point(s) of contact for the patient for any related functional or content question along with multiple communication methods (phone, email, etc.).
- They need information on how patient identity and health information, including sensitive information, is secured or shared by using the digital ID. This includes knowing about AI use or if AI is providing health information/monitoring/diagnostics. (Light Collective - No Aggregation Without Representation).

PR-9 b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

Use guidelines that safeguard sensitive information such as those developed for HIV, mental health, and substance use disorders.

E. Technology Vendors, Data Providers, and Networks

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

Continue to invest in HL7 FHIR API technologies. These play the single largest role in the short-term in bringing healthcare data into the modern web standards, lowering the bar for development with healthcare data.

3. Technical Standards and Certification

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A provider directory is a reference file with several actors updating their information. No private organization is incented to produce such a clean, rich resource, and CMS could play the role in publishing this directory. This is a common good, and free licenses could be granted or if put into a product, it could be licensed. It is a necessary reference for many players and has caused great inefficiencies in the current healthcare environment. It needs concerted efforts to establish portals with one and done entry until the next change is in order. Maintenance and upkeep are required if there is to be sustainability in the updating of information.

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

b. What would be the drawbacks?

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

The benefits would be in placing emphasis on data exchange via modern web standards, accelerating data model harmonization, new entrants into this space, empowered patients, expanded data for research, and lowering of administrative overhead.

Some drawbacks could include opportunity costs for exploration of other innovative technologies, such as AI. But more likely is the ongoing expansion of the digital divide, where providers with limited technological capabilities may continue to fall further behind.

4. Data Exchange

TD-15. Regarding bulk FHIR APIs:

TD-15 a. How would increased use of bulk FHIR improve use cases and data flow?

TD-15 b. What are the potential disadvantages of their use?

We view bulk FHIR as a technology that holds a lot of promise, especially for population health needs such as tracking immunization data. We hope CMS can help prioritize these technologies. Much work needs to continue to be done for routine patient care through other FHIR implementation guides, so perhaps bulk FHIR is not the top priority today.

F. Value-Based Care Organizations

1. Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

Patients aren't necessarily aware of the alternative payment model (APM) they are participating in. The APM partners as a governing body should collaborate. The technology and data standards exist; successful collaborations require actionable business cases, shared governance, and operational best practices to achieve data exchange. Collaboration on use-case, problem-solving, sharing knowledge, and organization pilot work is what moves adoption. There is a general lack of digital products that meet the needs of the organizations looking to invest limited resources in expanding their technological capacity. Most products exist to serve individual bespoke use cases or to meet regulatory requirements. They are rarely developed with patient or end user input, or with long term quality improvement goals in mind. This is somewhat due to the difficulty in predicting the future state, especially when target deadlines for base requirements for digital adoption keep getting pushed back.

A good example of collaboration with a regional HIE and community-based organizations (which could be a use case for an ACO to adopt) is the Bronx Regional Health Information Organization (RHIO) membership, where over 400 data provider organizations generated a community-based organization face sheet to be shared with social service site staff to operate more efficiently with access to data through the RHIO. The reports speed up client intake with demographic prefilling, support care transition and care management by providing patient history, and inform supportive services needed by clients. Data are securely shared based on existing privacy standards on a minimal need to know basis. The RHIO's data eliminates logging into multiple portals and feeds the data in a structured format. The RHIO and a Settlement House developed the face sheet concept, and the output produces risk scores in the face sheet along with stratified population-based aggregated reporting. It looks at unplanned readmission in 30 days, predicts if the patient will have unplanned inpatient utilization, classifies the patient into a patient cohort risk group, predicts if the patient will be an emergency department (ED) high utilizer, and classifies that risk.